

Exercise URL: <https://github.com/djhshih/exercise-wes>

Question1. Build a minimalistic pipeline to align the read pairs of tumour and normal samples to GRCh38 and identify somatic mutations in the tumour. Write good quality Bash scripts that can be re-used. Minimize disk IO. (Hint: There are two real somatic mutations in this synthetic dataset.)

Sol:

Please see the pipeline at **WES_analysis_pipeline.sh**

Example command to run the pipeline: **Example_task.sh**

For the environment configuration of the pipeline, please see **env_config_for_wes.sh** and **wes_env.yml**.

Question2. Describe how your pipeline can be modified to produce high quality mutation calls. Describe key steps that need to be added to the pipeline so that it can be used in production.

Sol:

To produce high quality mutation calls, firstly, the quality control and trimming of raw sequencing data of samples can be conducted by fastp before read alignment. Secondly, to further improve the alignment quality, local realignment of insertions and deletions by GATK IndelRealigner and the base quality score recalibration (BQSR) by GATK BaseRecalibrator can be performed on BAM files. Thirdly, somatic mutation with higher confidence can be further obtained by comparing the mutation calling results of different somatic calling tools, such as VarsCan and GATK MuTect2 (Variant calling by GATK MuTect2 and variant filtering calling by GATK FilterMutectCalls). Finally, the mutation can be further annotated by tools such as ANNOVAR.

To make the pipeline suitable in production, there are some key steps needed to be added:

1. Check before Running:

The pipeline would enable the user to test its reliability and potential problems by dry run. The pipeline would be able to create the output path if the path does not exist.

2. Logging and Monitoring:

Implement logging throughout the pipeline to record important information, the running task progress, errors, and warnings. When encountering error, indicate the location of its occurrence for troubleshooting. The execution and resource usage of the pipeline would also be able to be monitored.

3. Pipeline Orchestration:

Utilize orchestration tools (e.g., Nextflow, Snakemake, cwl, wdl) to manage the pipeline execution.

4. Reentrancy for Breakpoint Executing:

Implement strategy to make the pipeline able to continue where it left off if interrupted, without the need to restart from the beginning.

5. Resource and Time Management:

Incorporate resource management strategies to maximize the use of computing resources and prevent exceeding of resource limits. Enable the user to set the computing resources and running time for the task.

6. Parallelization:

Optimize the pipeline for parallel processing to enhance execution efficiency. The pipeline would support the job submission onto multiple nodes.

7. File Dependency:

Implement the update of associated downstream files when a dependency is updated.

8. File Protection and Deletion:

Protect important files produced by modifying permissions of to avoid accidental deletion. Mark the intermediate file as temporary file, which will be deleted after the running.

9. Documentation:

The pipeline should provide explanatory documentation for users, including instructions on how to set up and execute the pipeline, software versions, and document dependencies.

10. Notification

The pipeline will be able to notify users via email or other measures when a task starts or ends or encounters an error interruption.

11. Report Generation and Data Liberation

When the task is finished, the pipeline would be able to generate a result report to summarize and visualize the output results for the user. The data should be able to uploaded to the database or disk customized automatically.

12. Further Analysis and Statistics on WES data

Add more analysis and statistics steps in the pipeline to make it suitable for clinical application, such as:

1. Susceptibility gene identification
2. Statistics and pathway enrichment analysis of significantly mutated genes
3. NMF mutation signature analysis
4. Circos plot of somatic variations
5. Drug target analysis